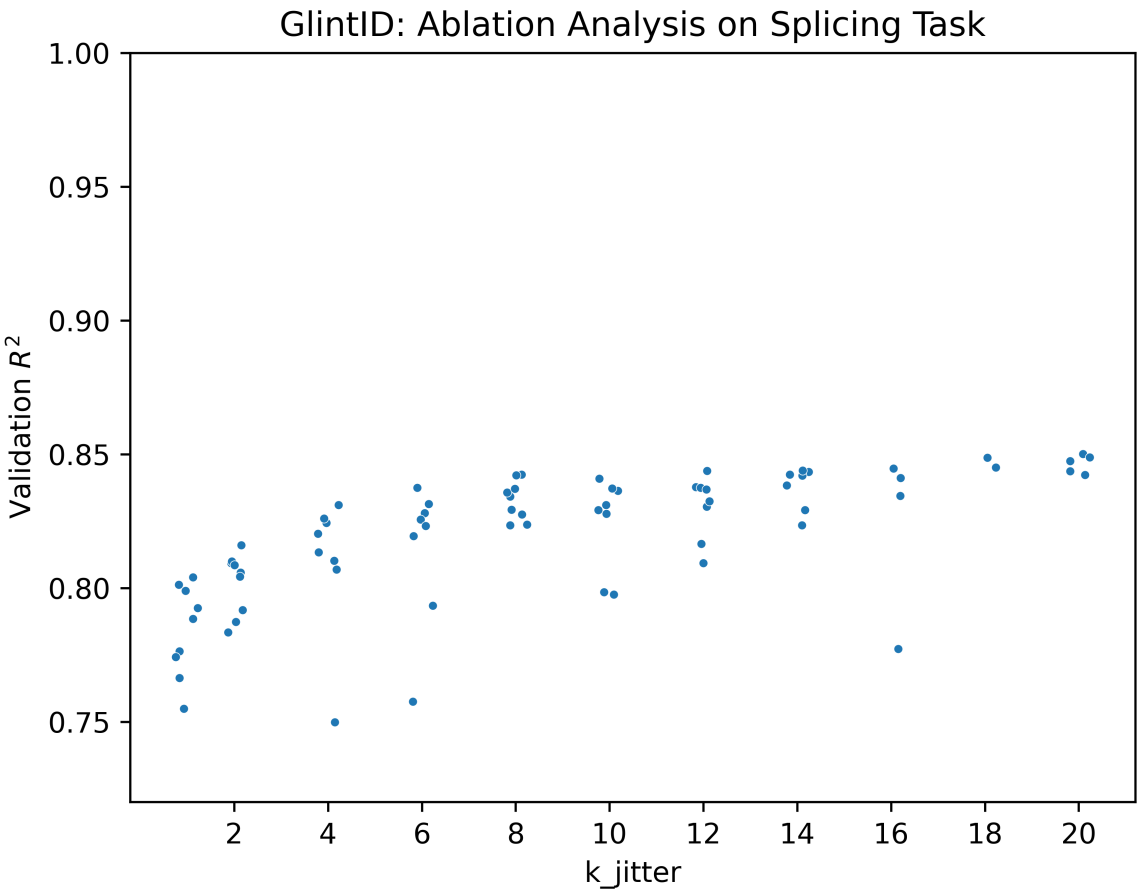

Figure R1. Validation performance of GlintID models trained on the splicing task with different seeds for various k during top-k selection for interaction terms.

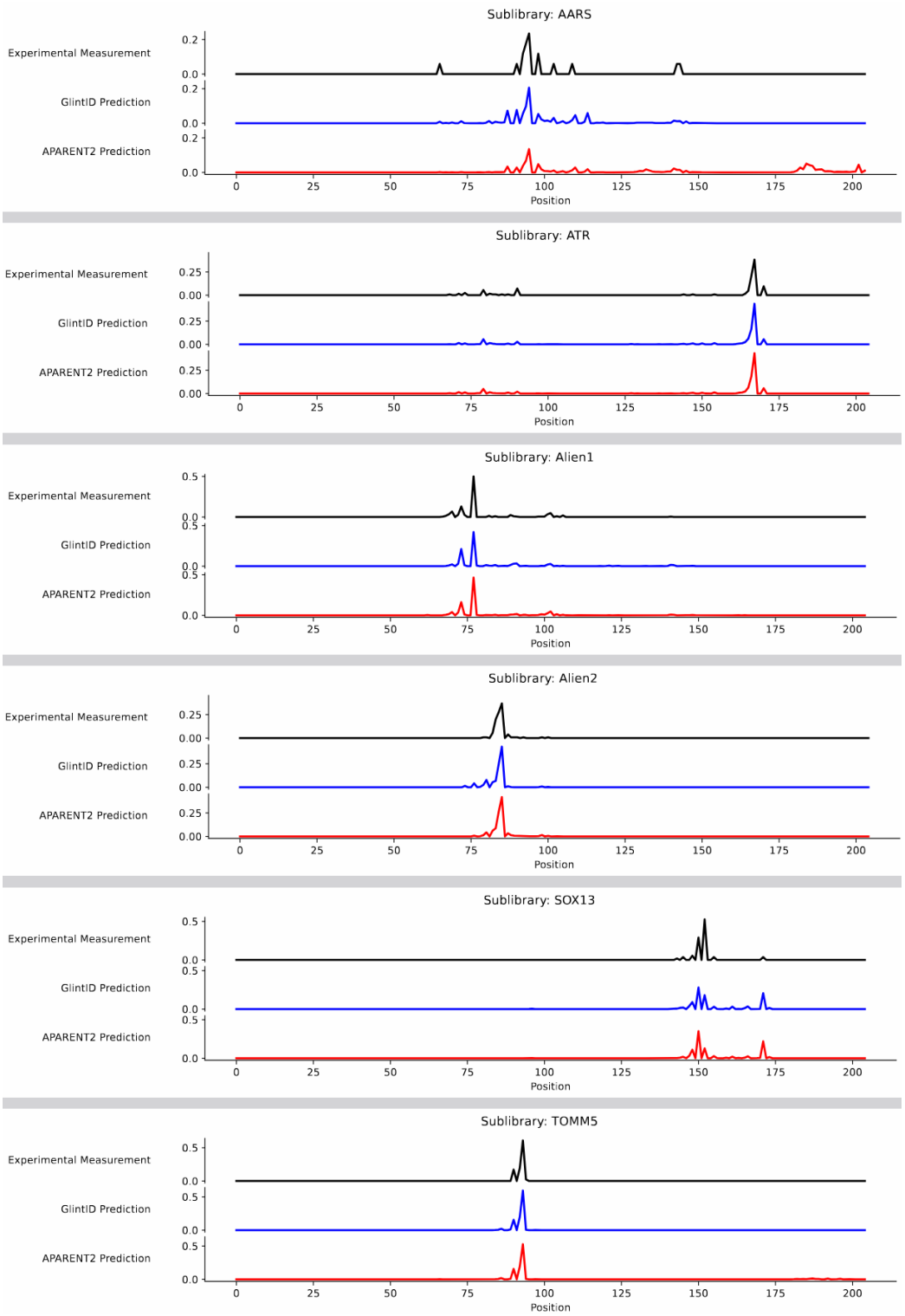

Figure R2. Cleavage track predictions of GlintID model trained on the polyadenylation track task. Comparisons of experimentally measured cleavage profiles, GlintID predictions, and APARENT2 predictions are shown for randomly sampled, representative examples from each withheld test sequence sublibrary. Each measurement/prediction track represents the probability distribution of cleavage occurring at a particular position along the given sequence.

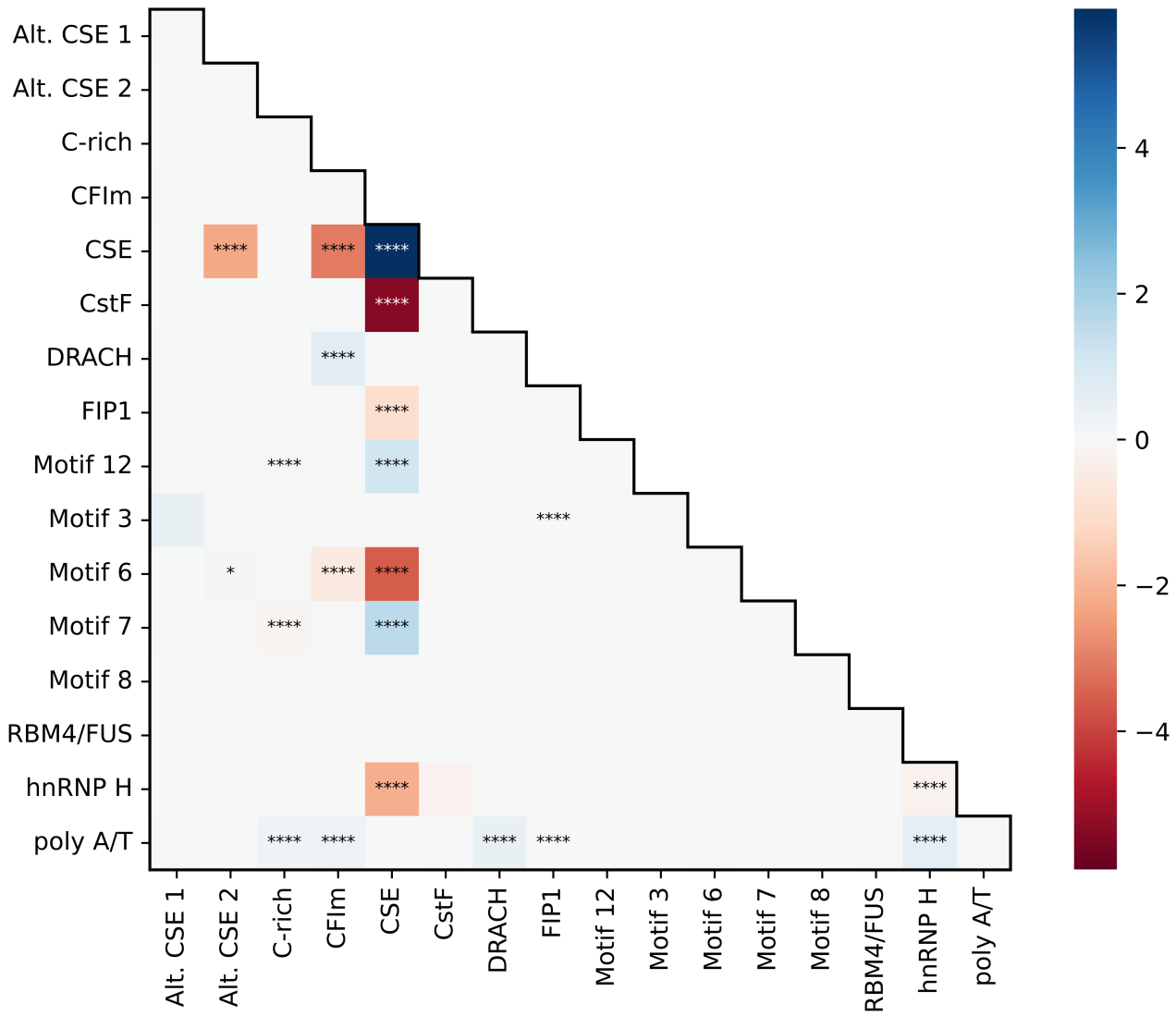

Figure R3. Interpretation of GlintID model trained on the polyadenylation task. Heatmap summarizing GlintID's learned interactions. See Appendix for details on interaction score computation. One-tailed paired sample t-test (Test C): *: $P \leq 0.05$; **: $P \leq 0.01$; ***: $P \leq 0.001$; ****: $P \leq 0.0001$. All P-values are Bonferroni corrected.

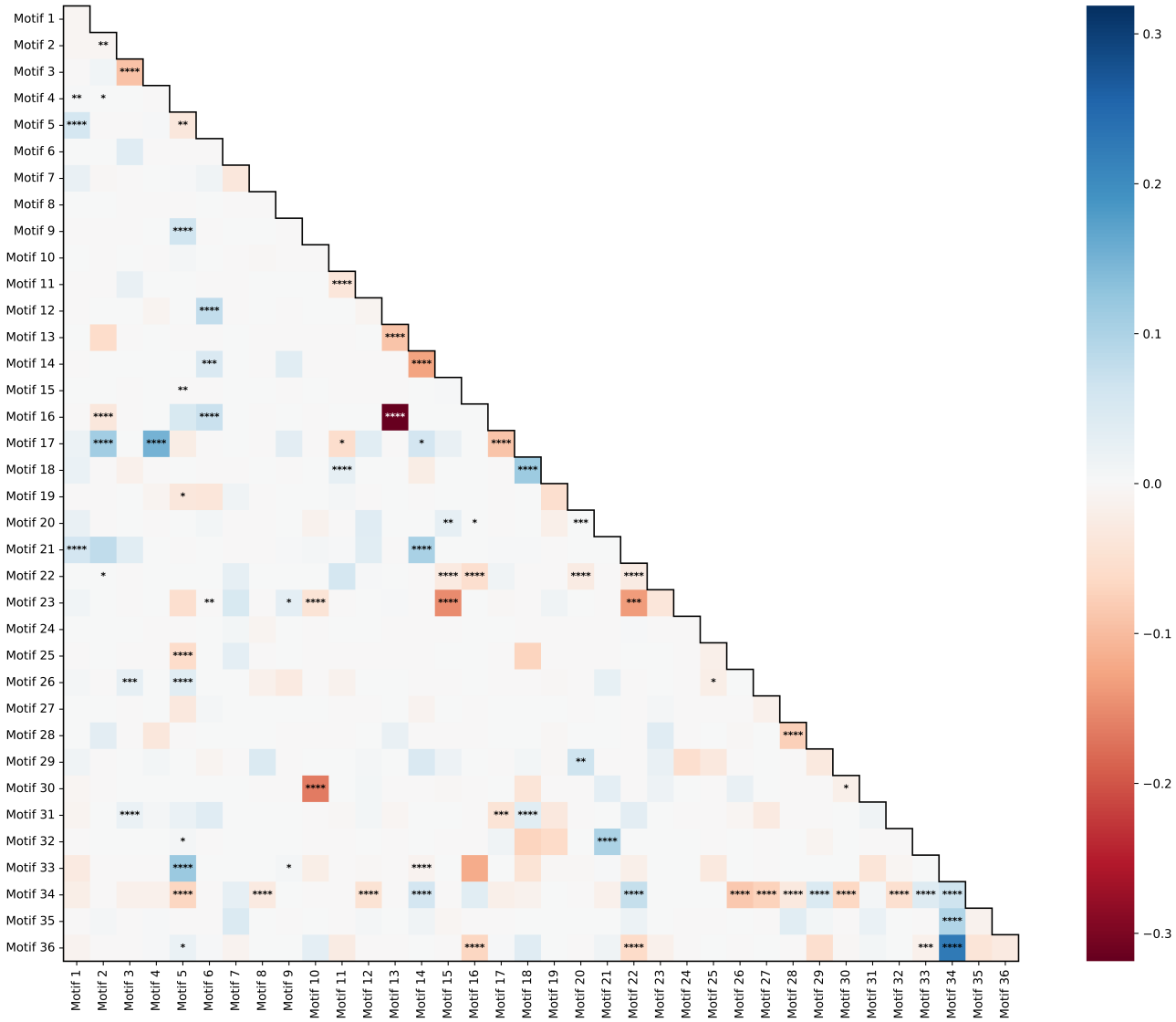

Figure R4. Interpretation of GlintID model trained on the splicing task. Heatmap summarizing GlintID's learned interactions. See Appendix for details on interaction score computation. One-tailed paired sample t-test (Test C): *: $P \leq 0.05$; **: $P \leq 0.01$; ***: $P \leq 0.001$; ****: $P \leq 0.0001$. All P-values are Bonferroni corrected.